

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

Here are 5 common desktop PC hardware issues along with their typical symptoms explained in simple terms:

1. Loose or Failing RAM (Memory)

- **Typical Symptom:** The PC turns on and fan lights light up, but the screen stays completely blank, often accompanied by a series of repeating **beep codes** from the motherboard or continuous restarting loops before Windows can load.

2. Power Supply Unit (PSU) Failure

- **Typical Symptom:** You press the power button and **absolutely nothing happens**—no fans spin, no lights turn on, and there is complete silence.

3. CPU or GPU Overheating

- **Typical Symptom:** The computer turns on and works fine initially, but **suddenly shuts off without warning** after a few minutes of gaming or heavy work, usually with the fans running at maximum, loud speed right before turning off.

4. Storage Drive Failure (HDD/SSD)

- **Typical Symptom:** The system runs extremely slowly, frequently freezes when opening files, or boots up to a black screen displaying an error like **"No Bootable Device Found"** or **"Operating System Not Found."**

5. Graphics Card (GPU) Fault

- **Typical Symptom:** Weird colored lines, dots, flickering patterns, or stretched pixels (known as **artifacts**) appear across the screen, or the display cuts out completely whenever you try to run a game or video editor.

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

1. Check the Outlet & Wall Connection: Rule out external power issues first.

- **What to check:** Unplug the charger and plug a lamp or phone charger into the same wall outlet to verify it is actively providing power. Make sure the charger cable is firmly seated into the wall adapter/brick.
- **Why:** Up to 20% of "dead charger" complaints are simply tripped circuit breakers, loose wall plugs, or bad extension cords.

2. Inspect the Cable & Adapter Brick: Look for physical wear or damage.

- **What to check:** Run your fingers along the entire length of the cable to feel for kinks, exposed wires, or severe bends. Feel the adapter brick—it should be slightly warm when plugged in. Check if the LED indicator on the brick (if present) is lit.
- **Why:** Frayed wires cause short circuits, and a completely cold brick usually means the internal fuse or component inside the charger has blown.

3. Examine the Laptop's Charging Port: Check for loose fits or debris.

- **What to check:** Gently wiggle the charging connector inside the laptop's port (DC jack or USB-C). It should fit snugly. Look inside the port using a flashlight for lint, dust, or bent pins.
- **Why:** A loose connector means the power pin isn't making contact. Dust buildup in USB-C ports often blocks the connector from plugging in all the way.

4. Perform a Hard Power Reset: Clear residual static electricity from the motherboard.

- **What to check:** Unplug the charger, hold down the laptop's power button for **30 seconds**, then release it. Plug the charger back in and attempt to turn it on.
- **Why:** Laptops can get stuck in a protective state due to static electricity buildup, which freezes the internal power controller chip until residual power is drained.

5. Perform a Battery Isolation Test: Determine if the issue is the battery or the charger.

- **What to check:** If your laptop has a removable battery, take it out. Plug the charger directly into the laptop and try turning it on without the battery installed.
- **Why:** If the laptop powers on normally without the battery, the charger and port are fine—the battery itself has failed and needs replacement. If it still won't turn on, the issue is the charger or the motherboard.

- 3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).**

Here is the step-by-step test procedure and the recorded voltage result for a standard 5V USB charger.

1. How the Measurement is Done

- **Multimeter Setup:** Turn the dial to **DC Voltage** (V $\overline{\text{---}}$) set to a low range (e.g., 20V DC).
- **Probe Placement:** Plug a USB cable into the charger. Touch the **Red probe** to the +5V pin (VBUS) and the **Black probe** to Ground (GND or the outer metal shell).
- **Inline Tester Alternative:** A USB power meter plugs directly into the charger port to read live voltage without exposed wires.



Common PC
Hardware Issues and !

2. Test Results

Parameter	Spec / Measured Value
Tested Device	Standard 5V USB Wall Charger
Recorded Voltage	5.08 V
Target Nominal Output	5.00 V
Acceptable USB Range	4.75 V to 5.25 V ($\pm 5\%$)

3. Evaluation & Conclusion

- **Status: PASS — Within Expected Range**
- **Explanation:** The measured voltage of **5.08 V** is safely inside the standard USB specification limit of **4.75 V to 5.25 V**.
- **Why it's slightly over 5.00V:** Most healthily functioning chargers output between 5.05 V and 5.10 V under no load. This slight buffer prevents the voltage from dropping below 4.75 V once a phone is plugged in and drawing active power.

4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

Hint:
Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

Here is how to safely test the +12V rail on a 24-pin ATX power supply and diagnose what low voltage means for your PC.

1. Step-by-Step Testing Guide

To test a power supply (PSU) when it is disconnected from the motherboard, you must use the "**Paperclip Trick**" to simulate the motherboard's power-on signal so the PSU turns on.

. Unplug & Isolate the PSU

Disconnect all power from PC components

- Turn off the switch on the back of the PSU and unplug it from the wall.
- Unplug the 24-pin main motherboard connector, CPU power cables, and GPU power cables from all internal components.

. Jump the PS_ON Pin (Paperclip Trick)

Simulate motherboard power-on command

- On the main 24-pin ATX connector, find the **Green wire** (PS_ON) and any **Black wire** (Ground).
- Insert a metal paperclip into the green pin hole and the other end into an adjacent black pin hole.
- Plug the PSU back into the wall and flip its power switch **ON**. The PSU fan should start spinning.

. Configure the Multimeter

Select correct measuring mode

- Turn the multimeter dial to **DC Voltage** (V $\overline{\text{---}}$) set to the **20V** range.

. Probe the Yellow and Black Wires

Measure the 12V rail voltage

- Insert the **Black probe** into any pin slot with a **Black wire** (Ground).
- Insert the **Red probe** into any pin slot with a **Yellow wire** (+12V).

2. Test Results

Wire Color	Multimeter Probe	Expected Range	Recorded Voltage
Yellow Wire (+12V)	Red Probe	11.40V to 12.60V	11.20 V
Black Wire (GND)	Black Probe	Reference (0V)	0.00 V

3. What a Reading Below 11.5V Indicates

A reading of 11.20V is significantly below the ATX standard minimum tolerance limit (11.40V). This indicates:

- **Power Supply (PSU) Failure:** Internal components inside the power supply (such as degraded output capacitors or failing voltage regulators) are worn out and can no longer deliver stable power.
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- **Why the PC Randomly Shuts Down:** The +12V rail powers your most energy-demanding components—namely the **CPU** and **Graphics Card (GPU)**. When you open a game or intensive software, those components draw heavy power. Because the voltage is already sagging at 11.2V, it drops even further under load, triggering the motherboard's safety protection circuit to immediately shut down the PC to prevent hardware damage.

Conclusion: The power supply is faulty and needs to be replaced with a reliable, properly rated PSU.